

SUPHAN BURI, Thailand

WMO: 484250

Lat: 14.474N

Lon: 100.139E

Elev: 29

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	64.0	66.8	53.1	60.1	71.8	56.0	66.9	73.6	12.1	77.3	10.7	77.5	1.5	350	0.340

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.5	99.9	79.7	98.3	79.9	96.8	79.7	83.4	93.3	82.6	92.1	81.9	91.1	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	161.6	87.1	80.4	158.2	86.6	79.8	154.9	86.1	47.4	93.8	46.5	92.1	45.8	91.4	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.2	7.8	DB	60.4	102.4	3.6	1.6	57.8	103.6	55.7	104.5	53.7	105.4	51.1	106.5	(4)
(5)			WB	56.0	84.5	3.4	1.0	53.6	85.2	51.6	85.8	49.7	86.4	47.2	87.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.2	79.5	82.9	86.0	88.6	87.8	86.4	85.2	85.1	84.6	83.9	81.9	78.9	(6)	
	DBStd	4.12	3.91	3.45	3.32	2.86	2.89	1.96	2.10	1.86	2.00	2.06	3.16	4.29	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	12492	913	922	1117	1157	1170	1092	1091	1087	1038	1052	956	897	(10)	
	CDD65	7017	448	502	652	707	705	642	626	622	588	587	506	432	(11)	
	CDH74	82962	4325	5544	8113	9562	9277	8048	7450	7294	6803	6889	5486	4171	(12)	
	CDH80	38370	1697	2529	4151	5410	4934	3839	3251	3118	2843	2843	2180	1575	(13)	
(14) Wind	WSAvg	1.9	1.4	1.7	2.0	2.0	1.9	2.2	2.3	2.1	1.3	1.4	2.0	2.1	(14)	
(15) Precipitation	PrecAvg	48.7	0.3	0.5	1.2	3.3	7.3	4.4	4.3	5.8	12.2	7.8	1.3	0.5	(15)	
	PrecMax	62.0	3.4	1.7	5.6	8.9	13.2	7.8	7.8	9.3	19.2	15.0	3.4	2.8	(16)	
	PrecMin	32.1	0.0	0.0	0.0	0.2	2.5	1.6	2.1	1.5	5.9	2.0	0.0	0.0	(17)	
	PrecStd	8.3	0.9	0.7	1.5	2.6	3.8	1.9	1.9	2.4	4.4	3.4	1.2	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	96.3	99.8	102.5	102.0	98.1	97.2	96.4	95.6	94.3	94.0	93.4	(19)	
		MCWB	76.3	77.0	77.1	79.1	80.9	81.4	80.4	80.3	80.8	80.6	78.9	77.7	(20)	
	2%	DB	91.7	94.6	97.8	100.4	99.3	96.0	94.8	94.4	93.9	92.5	92.4	91.5	(21)	
		MCWB	76.2	77.5	78.2	79.4	81.2	80.6	79.7	79.8	80.5	80.1	78.8	77.1	(22)	
	5%	DB	90.2	93.1	96.3	98.9	97.3	94.3	93.1	92.7	92.4	91.3	90.8	89.8	(23)	
		MCWB	75.6	77.4	78.4	79.6	81.1	80.5	79.7	79.5	80.1	79.7	78.1	76.1	(24)	
	10%	DB	88.4	91.5	94.8	97.2	95.3	92.7	91.4	91.2	90.8	89.9	89.2	87.7	(25)	
		MCWB	74.8	77.0	78.4	79.6	80.8	80.2	79.3	79.0	79.7	79.3	77.4	74.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	81.8	82.4	83.6	84.5	83.9	82.6	82.8	83.6	82.7	82.5	80.6	(27)	
		MCDB	89.0	91.8	94.0	94.5	95.6	93.9	92.1	92.0	91.8	89.9	89.8	89.0	(28)	
	2%	WB	78.9	80.5	81.2	82.5	83.4	82.7	81.5	81.2	82.0	81.8	81.1	79.2	(29)	
		MCDB	87.7	90.5	92.1	92.5	94.0	92.5	90.7	90.6	90.3	89.0	88.3	87.3	(30)	
	5%	WB	77.9	79.6	80.6	81.8	82.5	81.7	80.7	80.4	81.2	81.1	80.2	78.1	(31)	
		MCDB	86.0	88.8	91.0	91.6	92.5	90.9	89.8	89.7	89.2	88.2	87.3	86.1	(32)	
	10%	WB	77.0	78.7	79.9	81.2	81.7	80.9	80.0	79.7	80.4	80.5	79.2	76.9	(33)	
		MCDB	84.2	87.3	89.4	90.9	91.3	89.4	88.5	88.4	88.1	87.5	85.9	84.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	17.8	17.5	17.5	15.0	13.5	12.9	13.1	12.8	11.9	13.6	16.3	(35)	
		MCDBR	18.5	18.6	19.3	19.1	17.5	15.6	15.2	15.1	14.9	13.4	14.9	17.2	(36)	
	5% WB	MCWBR	5.9	4.9	4.6	4.2	4.5	4.6	4.8	4.5	4.8	4.3	4.6	5.6	(37)	
		MCWBR	15.9	16.8	17.7	17.7	16.0	14.5	14.0	14.1	13.7	12.5	13.4	15.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.535	0.611	0.659	0.586	0.499	0.452	0.444	0.445	0.476	0.576	0.518	0.512	(40)	
		taud	1.950	1.800	1.721	1.928	2.231	2.359	2.384	2.368	2.273	1.948	2.079	2.033	(41)	
	Edn at Noon		241	229	222	239	257	266	269	272	265	235	245	244	(42)	
		Edh at Noon	57	69	77	62	45	39	39	40	44	59	50	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		1535	1675	1757	1910	1770	1661	1557	1544	1539	1514	1538	1503	(44)	
(45)	RadStd		90	67	126	102	133	104	92	107	91	90	92	63	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.08	+1.15	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	+0.49	+0.50	N/A	N/A	+171	+169

Nomenclature: See separate page